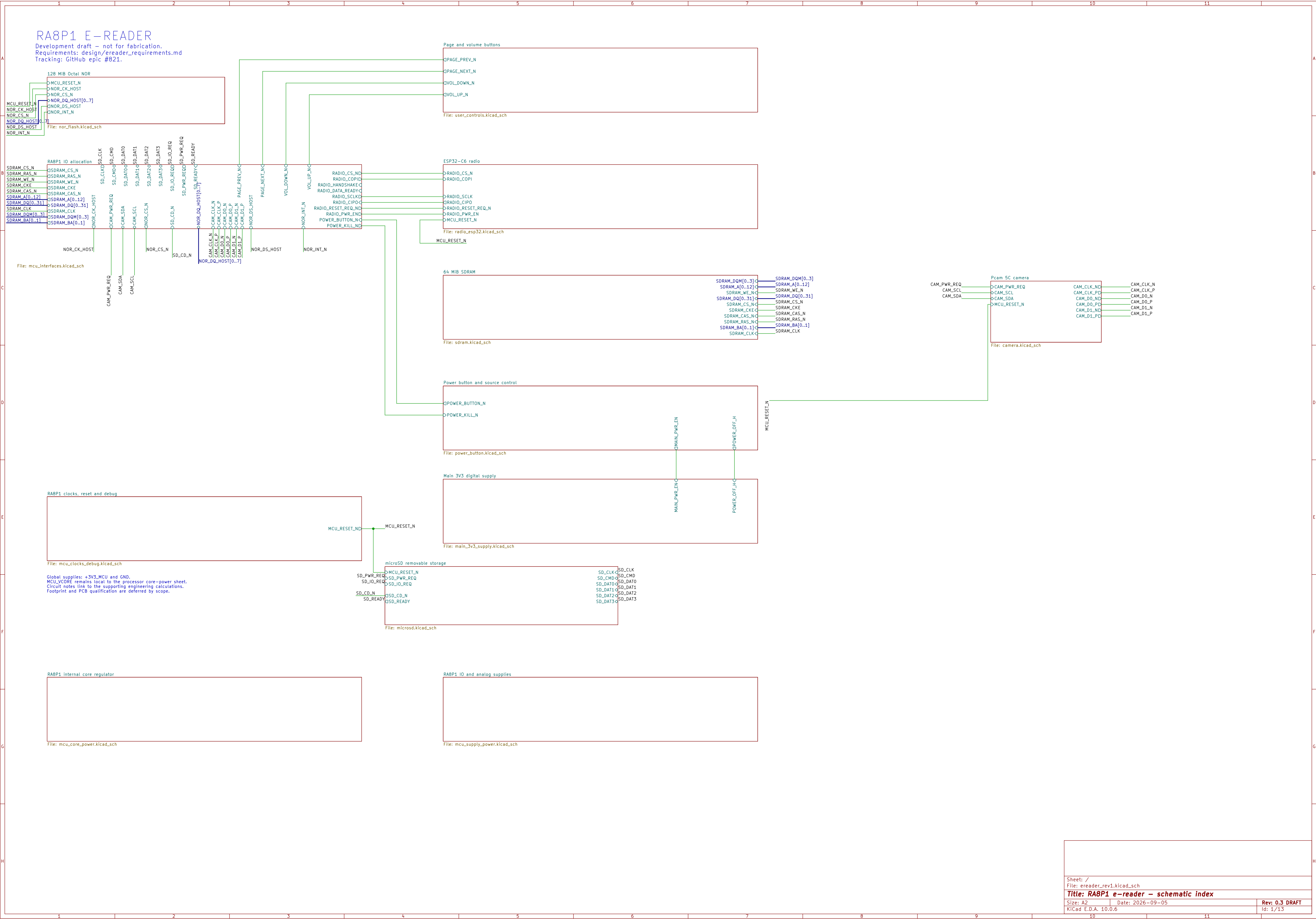
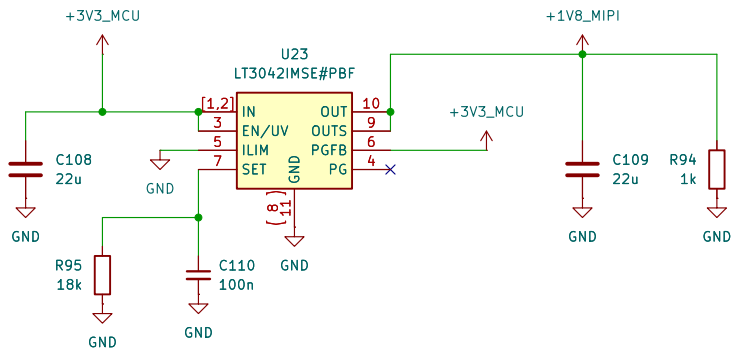


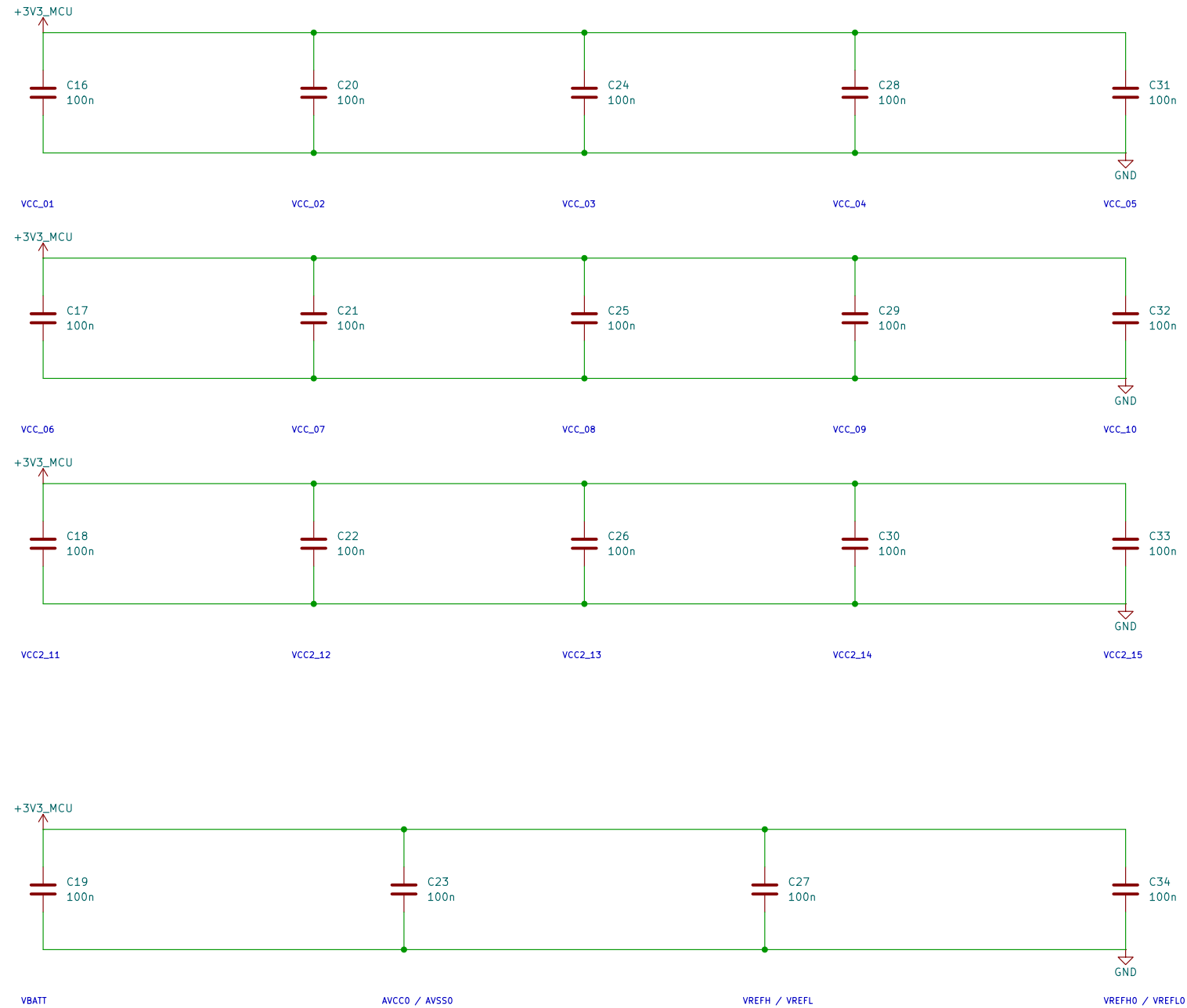
RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.



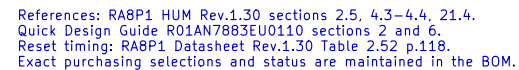
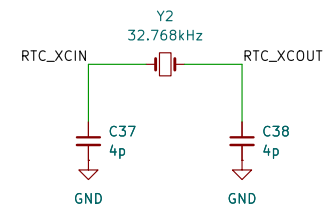
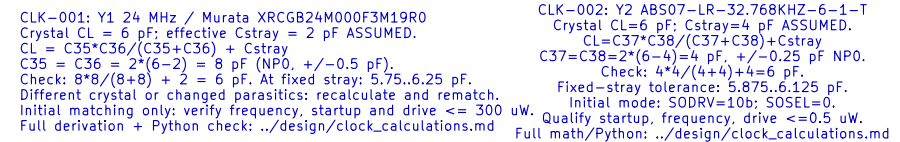


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

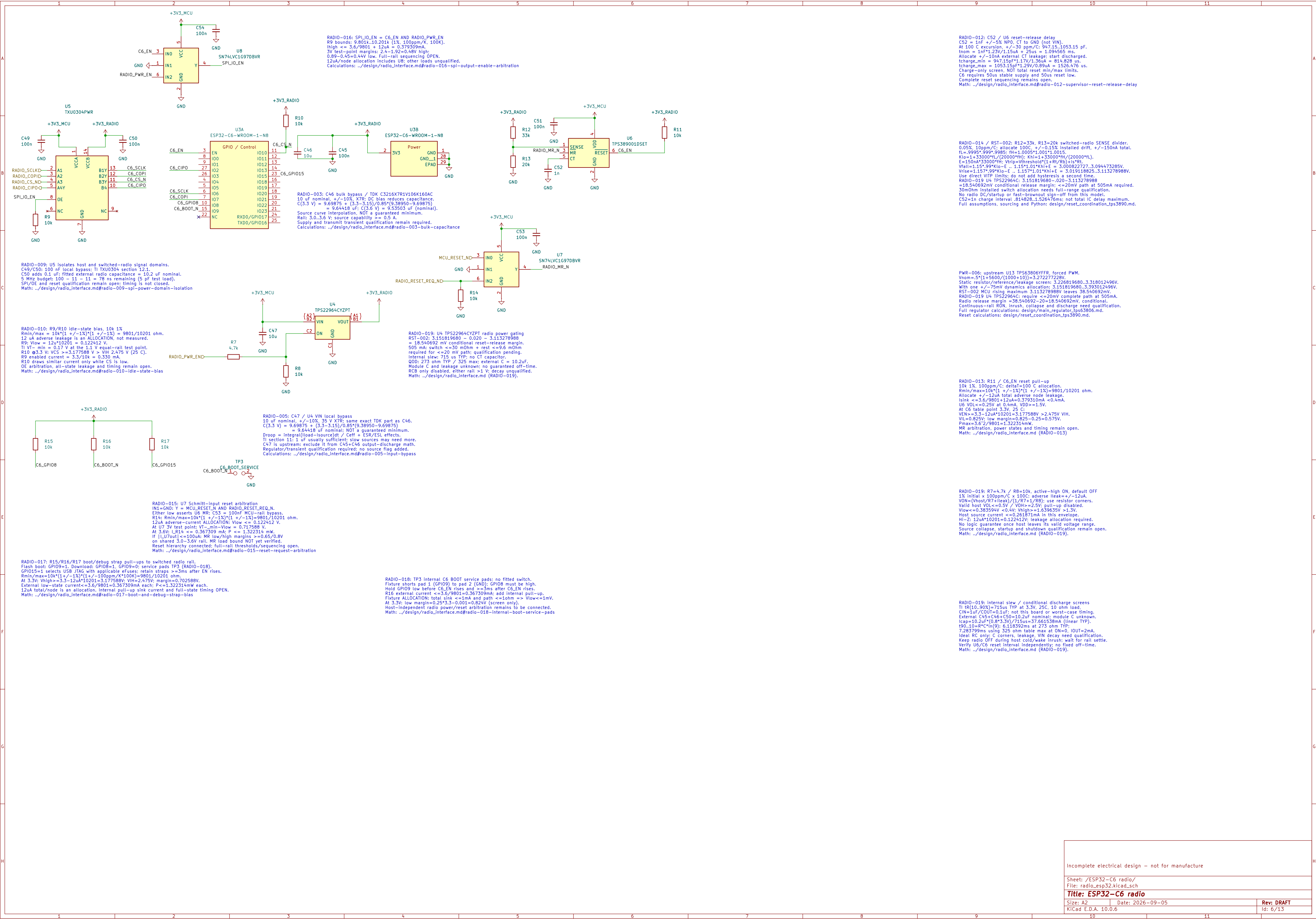


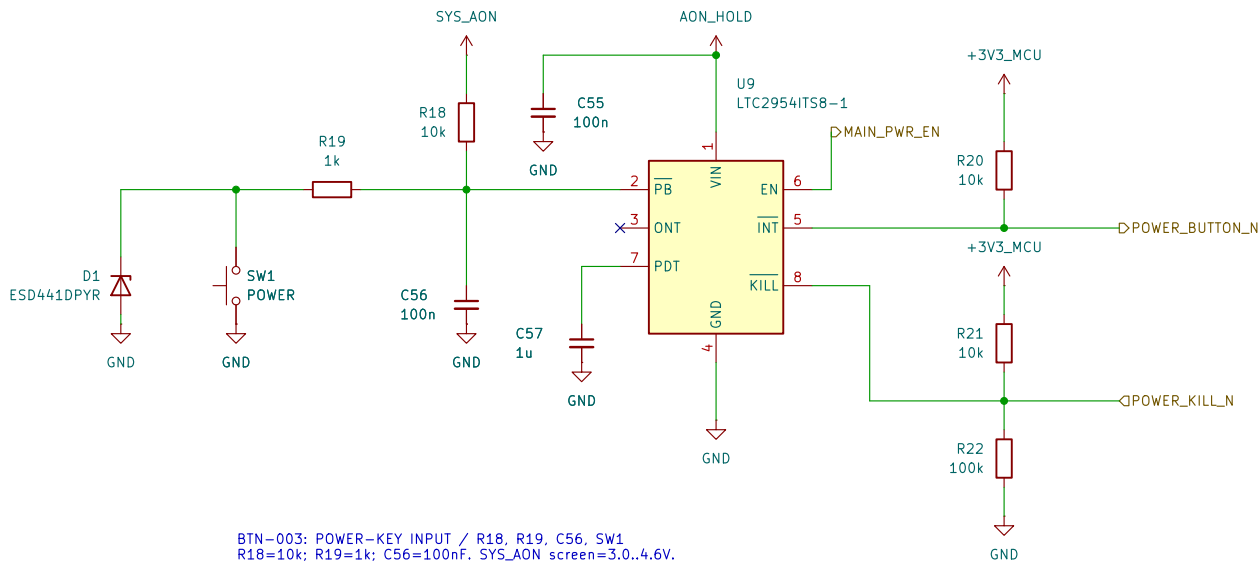
CMS-013: +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification: startup/inrush excluded.
AVCC/MPI remains within existing MCU allocation.
Main subunit: 2.075 + 0.020 = 2.095A; full system budget pending.
At 2.0496V, 3.2V input, 1.25W load, 75% efficiency:
input 2.961817158A; converter loss = 2.3694535727W
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage/interfaces.md.

RA8P1 e-reader		
Sheet: /RA8P1 IO and analog supplies/ File: mcu_supply_power.kicad_sch		
Title: RA8P1 I/O and analog supplies		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/13



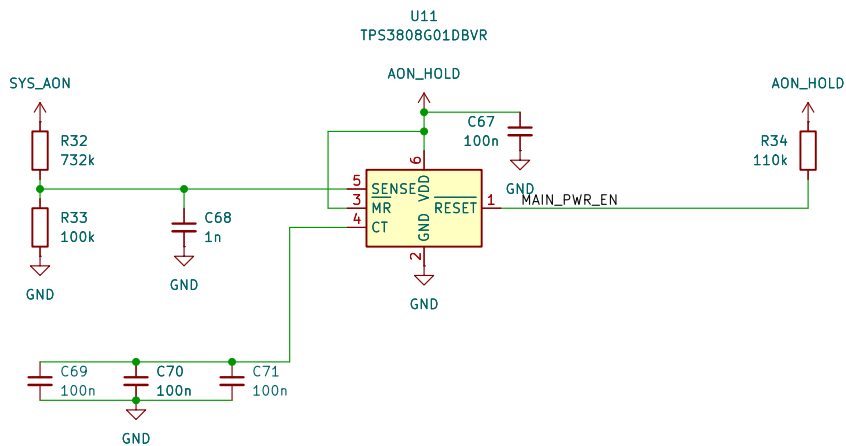
```
RST-002: U2 TPS3890U1DSET; R67=33k, R74=20k, R75=10k, C95=10n.  
Sense resistors: +0.05%, OhmCo over 100C, allocated + / -0.15% assembly/aging.  
Klo=9995.99*Klo-E = 9.9902E+25; E=1.0005E+10; Klo=33000.00*FfH; FfH=33000.00*FfH;  
Klo=4.133000*fH/(200000*fH); E=1.33000*fH/(20000*fH); E=150nA/330000*fH.  
Vfall=1.15*.99*Klo-E = 1.15*1.01*KHi+E = 3.00082727Z.3.094473285V.  
Vrise=1.157*.99*Klo-E = 1.157*1.01*KHi+E = 3.01911882S.3.113278988V.  
Main release margin=3.151819680-.3.113278988=.38.540692mV; no double hysteresis.  
C95: 10n COG +/- .5%, 30ppm/C over 100C; allocate + / -10nA external CT leakage.  
tcharge,min=10n*.95*.997*1.17/(1.35uA+10n)=1.842676ms > 2.4ms RES minimum.  
tcharge,max=10n*1.05*1.003*1.29/(90u-10n)=15.264758ms; NOT total-delay max.  
R75 MR pull-up: R=9801.10201ohm; allocate + / -50uA total adverse MR current.  
MR high margin=3.3*1.51819680-50uA/10201A=4.354595904V; sink <- .396190439mA.  
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.  
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.
```





BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(1020.1+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

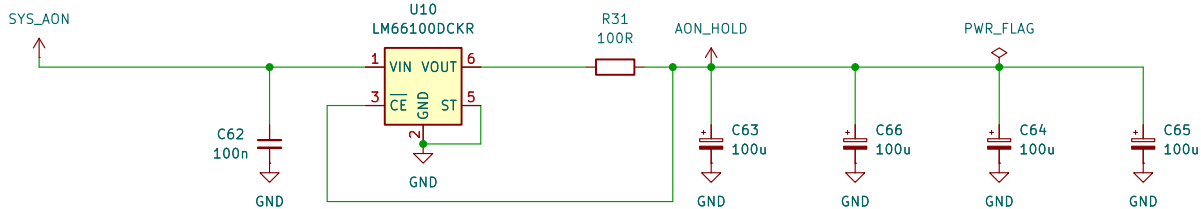
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF;
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / PWR-004: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% + 100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage (is a qualification allocation).
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Key-filter return allocation Ir=1.5mA; R=Rmax.
Total=10ms+R*C*ln((3.6-Ir*R)/(0.3-Ir*R)).
At 850uF: 34.647839ms; at 1mF: 38.997458ms <46.589403ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at 70C.
Qualify 0.2ohm FET, 10ms TOTAL response, derating and backfeed.
Gate/transient charge shares the existing 1uC reserve.
VBATT firmware/option invariant remains unimplemented; #846 unresolved.
Sources/Python: ../design/system_power_design.md SYS-007;
../design/main_regulator_ltc3119.md PWR-004.

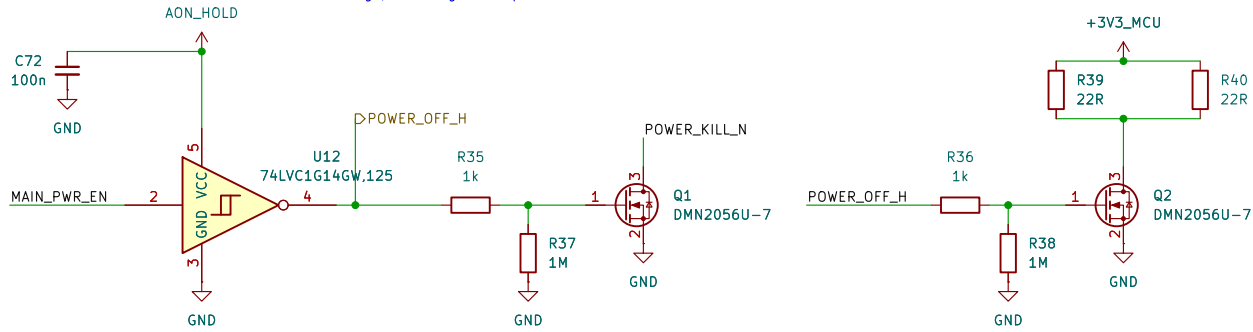
INCOMPLETE POWER CONTROL - NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission-latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR-006.
MCU reset: design/reset_coordination_tps3890.md RST-002.
Radio release margin and system shutdown coordination remain open (#846).

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



SYS-007 / PWR-004: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 100uF/10V; Cmin=400uF*0.9^3=291.6uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol=1.125mA incl 500uA bank leakage allocation.
Pre-trip reverse: 80mV/98.01ohm<0.817mA. Itotal=1.942mA.
Vinitial=3.263705831-max(0.250,0.001125*102.24)
=3.013705831V (threshold OR resistive-drop limit).
t_hold=[291.6uF*(3.013705831-2.7)V-1uC]/1.942mA
=46.589403ms >38.997458ms shutdown at 1mF.
Key-return-aware hold margin=7.591945ms at 1mF.
1uC covers initial undershoot + reverse/gate transitions.
Qualify VIN=0..4.6V, slow/floating-source collapse and load.
R31 short: 4.6^2/98.01<0.216W; 1W nominal rating.
Control only: no MCU/audio/display loads on AON_HOLD.
Separate unadopted eFuse extra load is not included.
Sources/Python: ../design/system_power_design.md SYS-007;
../design/main_regulator_ltc3119.md PWR-004.

SYS-007 / PWR-006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR-006.
Sources/Python: design/system_power_design.md SYS-007;
design/main_regulator_tps63806.md PWR-006. Reset coordination: #846.



Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RABP1 E-reader - power button and source control

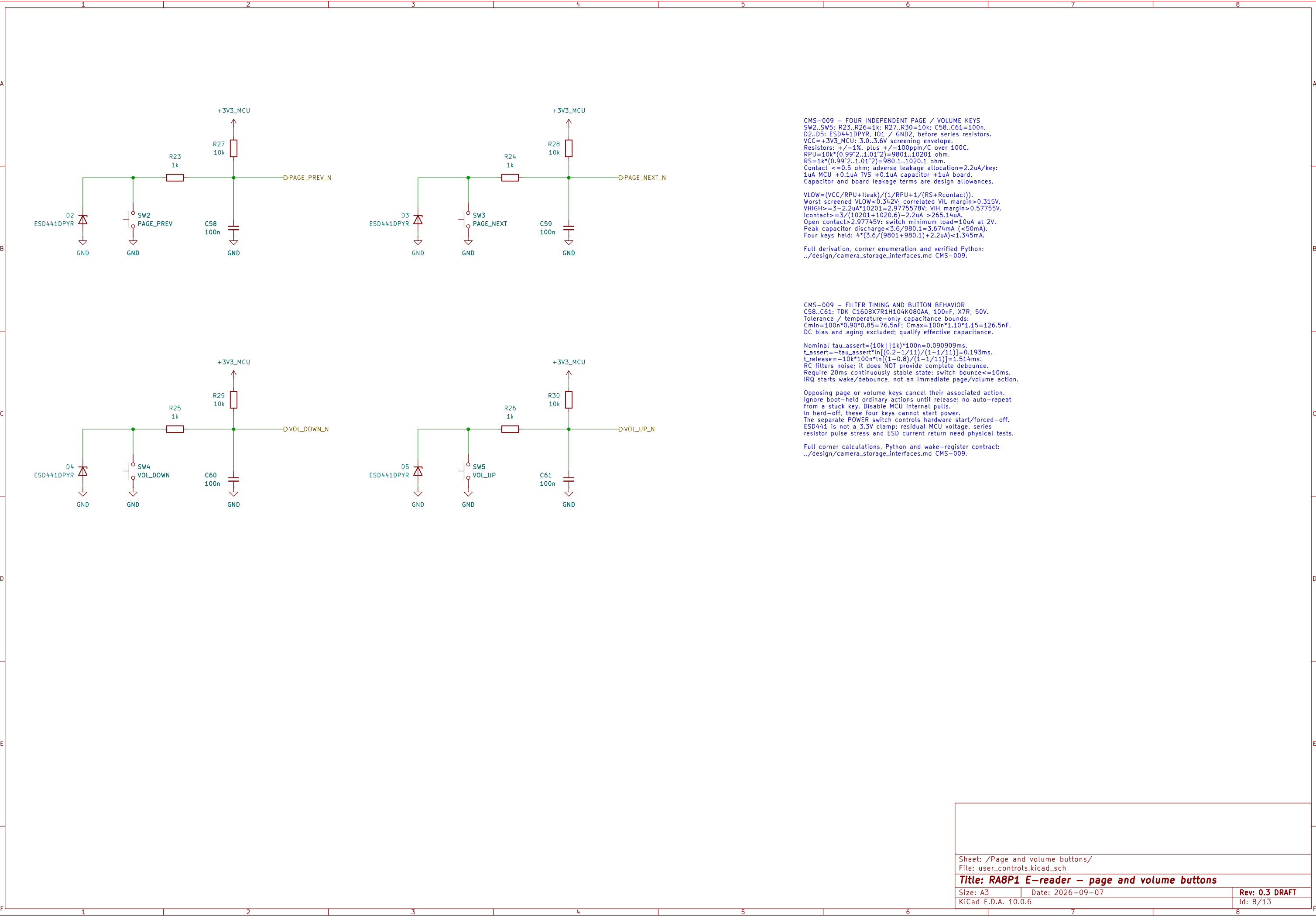
Size: A3

Date: 2026-09-07

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 7/13



CMS-009 - FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

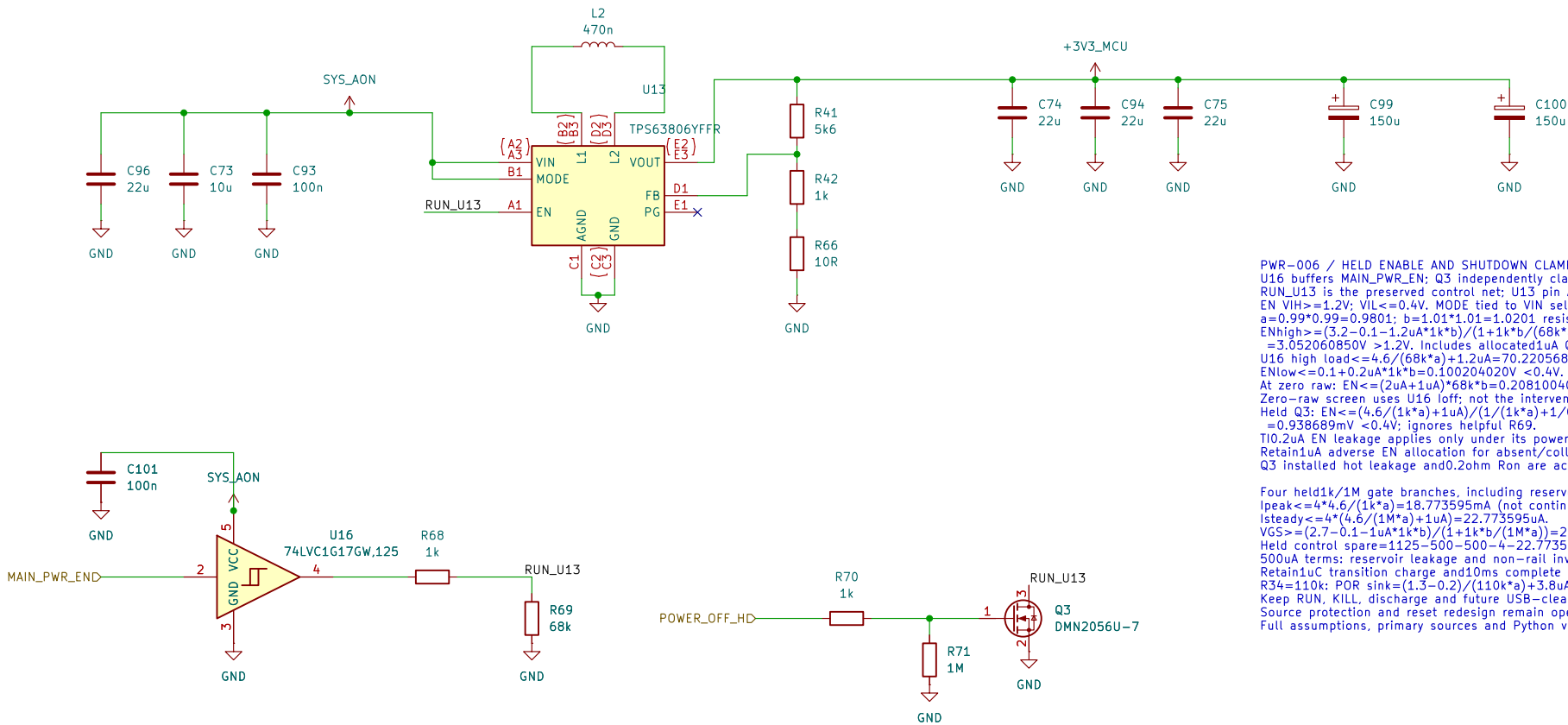
CMS-009 - FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



PWR-006 / HELD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWR_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
a=0.99*0.99=0.9801; b=1.01*1.01=1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b/(68k*a))
=3.052060850V >1.2V. Includes allocated1uA Q3 off leakage.
U16 high load<=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
TI0.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Ron are acceptance allocations.

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*4.6/(1k*a)=18.773595mA (not continuous rating).
Isteady<=4*(4.6/(1M*a)+1uA)=22.773595uA.
VGS>=(2.7-0.1-1uA*1k*b)/(1+1k*b/(1M*a))=2.596278V.
Held control spare=1125-500-500-4-22.773595=98.226405uA.
500uA terms: reservoir leakage and non-rail inverter-input allocation.
Retain1uC transition charge and10ms complete response allocation.
R54=110k; POR sink=(1.3-0.2)/(110k*a)+3.8uA=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Full assumptions, primary sources and Python verification: click this note.

PWR-006 / FEEDBACK AND LOCAL PASSIVES
Vnom=0.500*(1+5600/(1000+10))=3.272277228 V.
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed bounds: initial*TCR(100C)*aging*assembly, then ohms.
R41=5600*(1+/-0.0002)*(1+/-0.0005)*(1+/-0.001)
*(1+/-0.0005)+/-0.02 = 5587.669237..5612.349243 ohm.
R42=1000*(1+/-0.0001)*(1+/-0.0002)*(1+/-0.001)
*(1+/-0.0005)+/-0.01 = 998.190970..1001.810970 ohm.
R66=10*(1+/-0.01)*(1+/-0.02)*(1+/-0.01)+/-0.05
=9.554980..10.455020 ohm; one combined aging/assembly 1%.
Vlo=0.495*(1+R41min/(R42max+R66max))-100nA*R41max.
Vhi=0.505*(1+R41max/(R42min+R66min))+100nA*R41max.
Vstatic=3.226819680..3.318012496 V (FPWM only).
One combined +/-75mV regulation/ripple/transient/routing allowance:
Vcomplete=3.151819680..3.393012496 V; qualification required.
SDRAM static VOH margin=2.4-0.7*3.393012496=24.891253mV.
This is not measured SI/noise headroom or reset qualification.

L2: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
=>0.3708..0.5698uH versus TI0.37..0.57uH. Qualify hot/current.
C74/C75/C94: 3*22uF*0.8*0.85*0.6=26.928uF > TI21uF.
C96: 22uF*0.8*0.85*0.6=8.976uF > TI4uF input minimum.
0.6 is allocated combined bias/aging retention, NOT guaranteed.
Samsung typical output-bias retention at3.393012496V=90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LOAD, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold-start screen, not measured load.
Pout=3.393012496*1.8=6.107422493W.
At allocated75% efficiency: Pin=8.143229991W; Iin@3.2V=2.544759372A.
Loss=2.035807498W: NOT an acceptable continuous thermal result.
At90% efficiency: loss=0.678602499W.
TI screen=70C+78.8C/W*0.678602499W=123.474C.
TI board theta-JA is not the final PCB thermal resistance.
2.25A screen: Pin=3.393012496*2.25/0.75=10.179037489W.
Converter limit, source limit, startup and thermal envelope need validation.

Native storage inventory (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on adds10.2uF =>489.81uF.
C99/C100 remain150uF each, C94 adds22uF to the output.
Old bootstrap/private VCC/VC capacitors were not main-rail storage.
Whole-main ceiling remains1mF for shutdown; no LTC stability transfer.
Cout lower acceptance and distributed PDN/startup are separate checks.

Preserved discharge screen: Rmax11.4211ohm; key return<=1.5mA.
Vinf=1.5mA*11.4211ohm=0.01713165V.
t=1mF*11.4211ohm*ln((3.6-Vinf)/(0.3-Vinf))+10ms
=38.997458ms to0.3V, below held46.589403ms screen.
Response10ms includes command/clamp/discharge delays.
Held budget retains1uC transition charge and98.226405uA spare.
SYS_AON source protection, loaded startup and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions, sources and executable Python: click this note.

Sheet: /Main 3V3 digital supply/
File: main_3v3_supply.kicad_sch

Title: RA8P1 E-reader – main digital 3.3 V supply

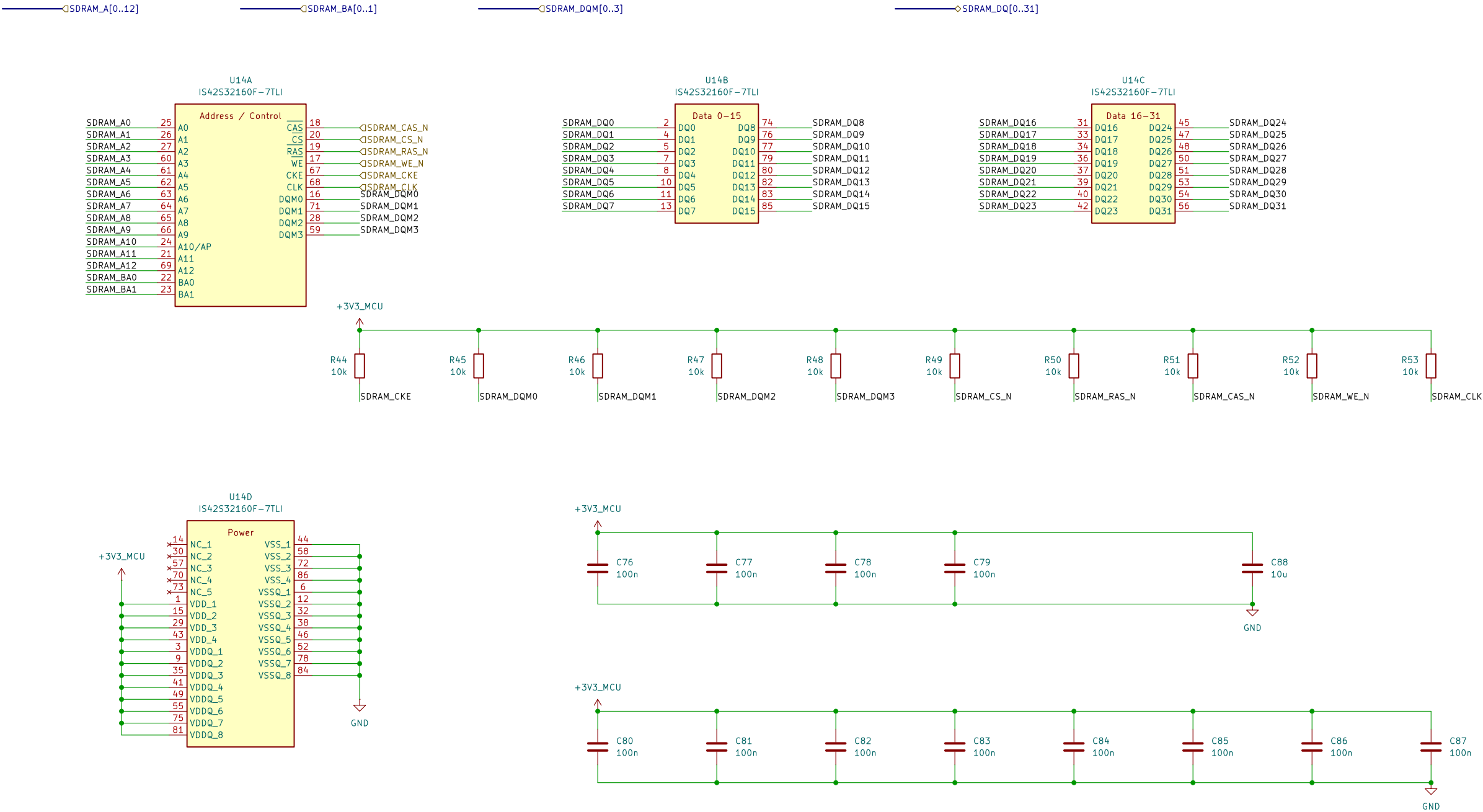
Size: A3 Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 9/13

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

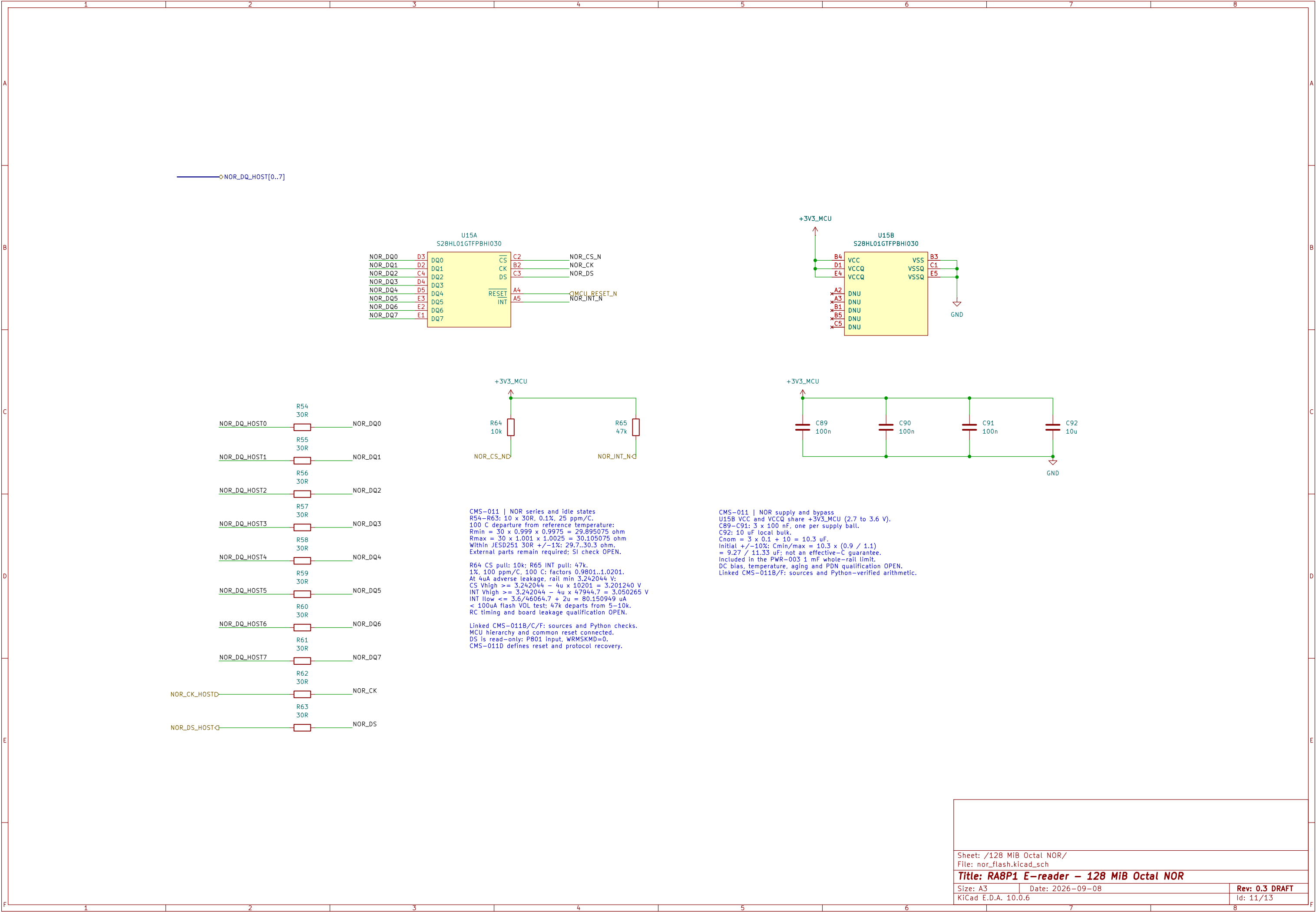
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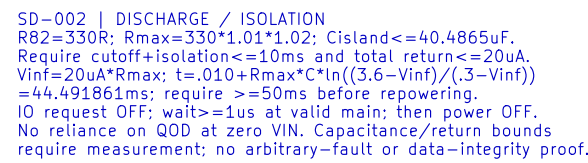
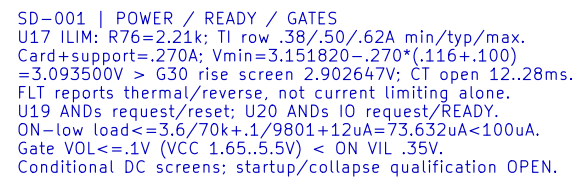
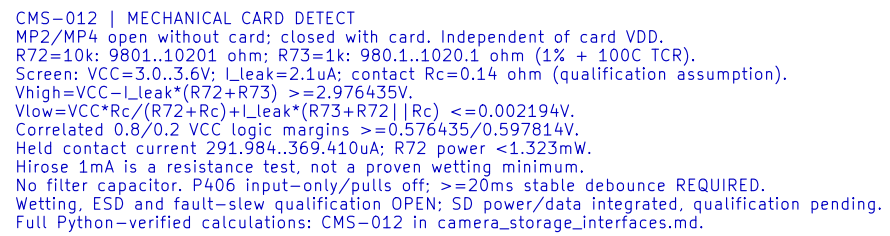
Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 10/13





SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=,750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
lin=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=,376190mA<,4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

Incomplete electrical design – not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

Title: RA8P1 e-reader – microSD removable storage

Size: A3

Date: 2026-09-08

Rev: 0.3 DRAFT

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